# Schooling drives neurocognitive development: Evidence from a longitudinal school cut-off design

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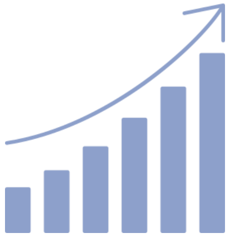
"Five-to-seven shift"

*(Sameroff & Haith, 1996)*



age-related changes
(maturation)

Massive changes in children's **neurocognitive functioning and brain development**

start of
formal schooling

= start of formal reading
and arithmetic instruction

# What do we know?

- Studies focusing on brain development during **ages 5 to 7** and linked to **reading and mathematics**?

- Most of this research has focused on older (primary school) children, after start of formal schooling

- Broadly distributed network, not yet lateralized in young children
  *(Benischek et al., 2020; Pugh et al., 2013; Raschle et al., 2012; Xiao et al., 2016; Skagenholt et al., 2018; Sokolowski et al., 2017)*

# What do we miss?

01

Very little longitudinal evidence during this age range

02

Longitudinal studies cannot isolate effects of schooling from age-related maturation

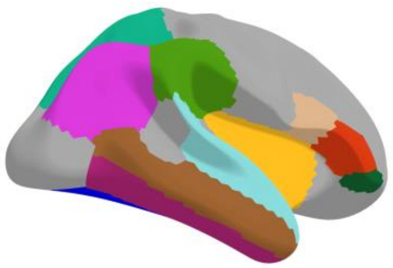
03

Number of studies in neurotypical children is limited

# Methods

Participant sample: 5-year-olds
(max. age range = 6 months)

Two timepoints of data collection,
one year in between.



**Structural MRI**: gray matter volume, cortical thickness, and cortical surface area of predefined ROIs

**Functional MRI**: word and number stimuli presented in passive target detection paradigm
*(Dehaene-Lambertz et al., 2018)*


**Diffusion-weighted MRI**: DTI data analyzed with AFQ *(Yeatman et al., 2012)* focusing on five major white matter tracts
*(Vandecruys et al., 2024, Dev Sci)*

**Behavioral data**: Mathematics, reading, and relevant precursors
*(Vandecruys et al., 2024, JEP)*

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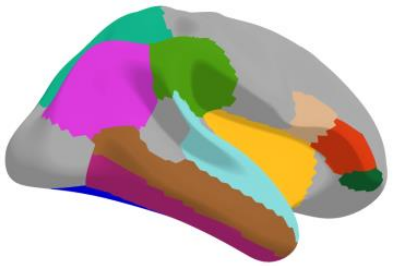

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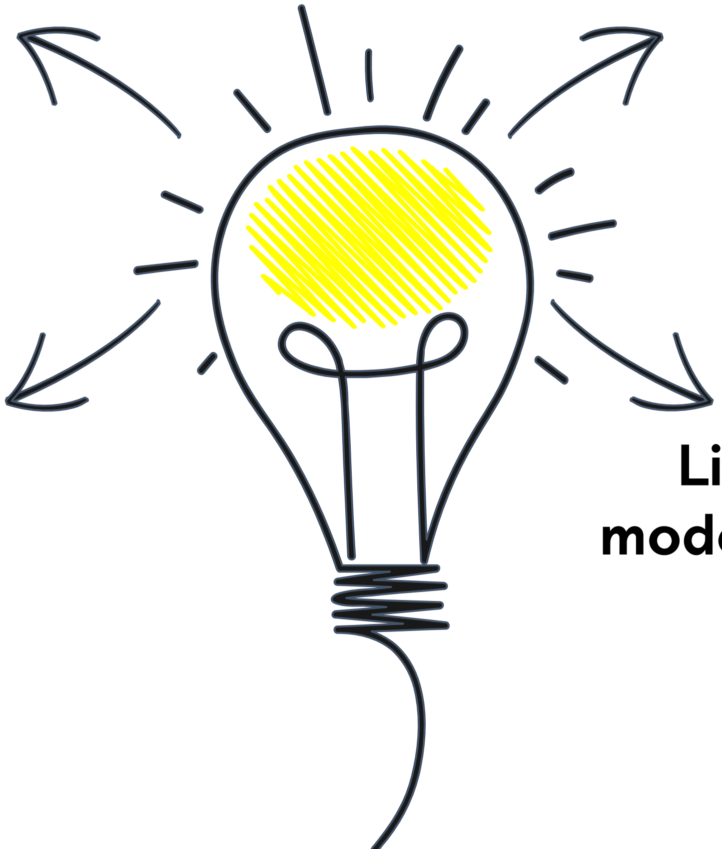
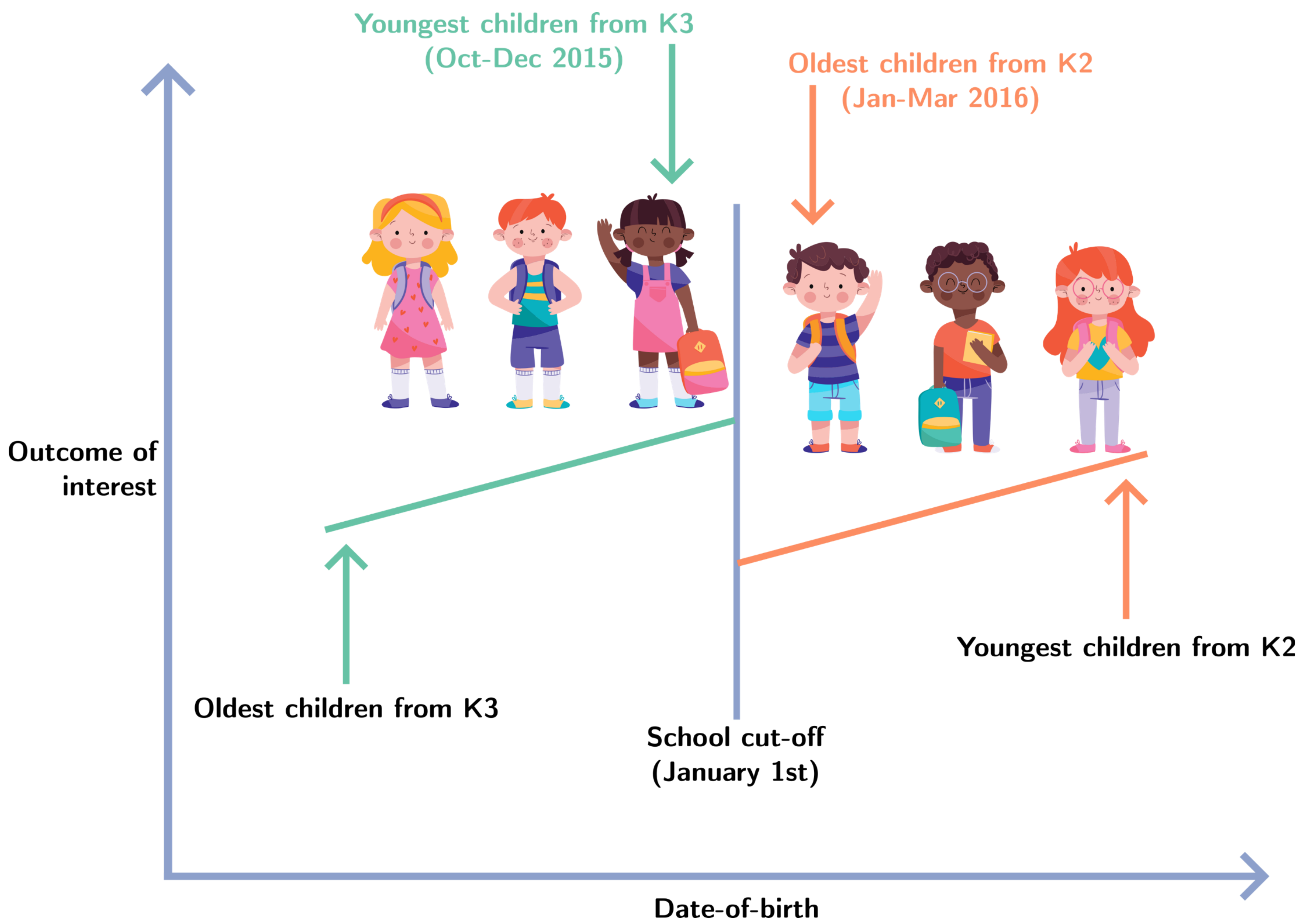


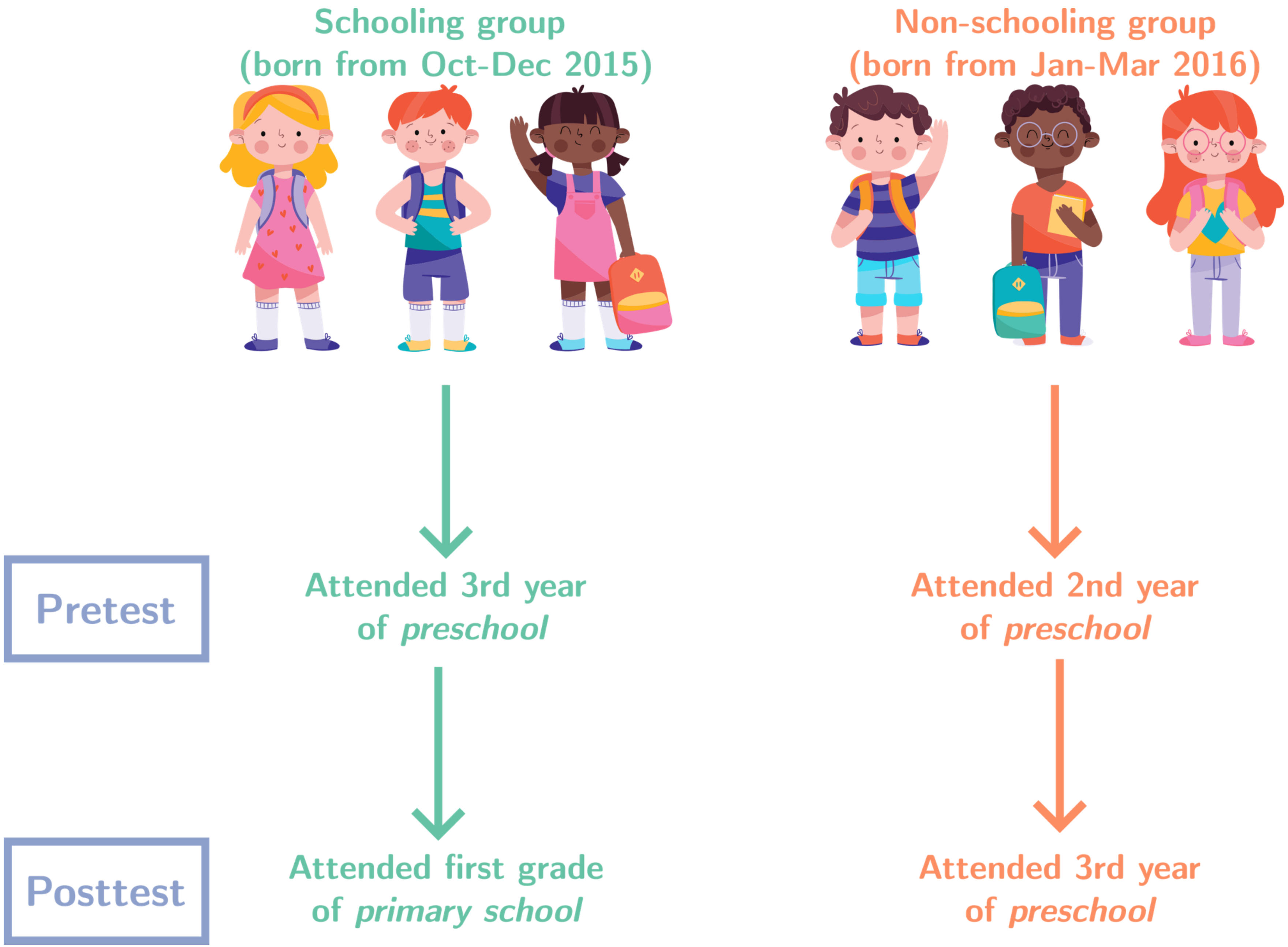
Longitudinal
**School cut-off design**

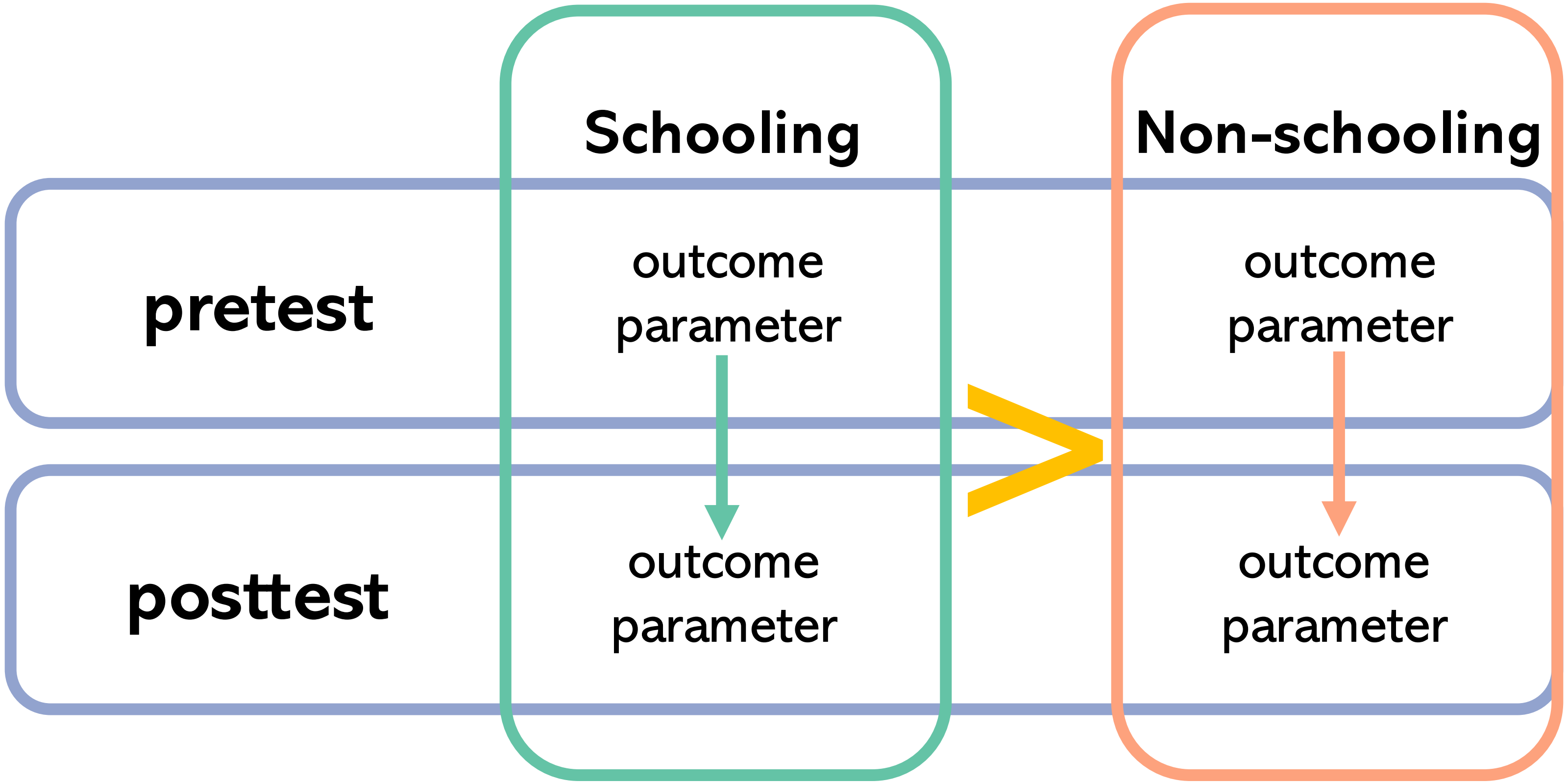
*(Morrison et al., 1995; 2019)*

**Linear mixed-effects models** (group x timepoint)



Youngest children from K3
(Oct-Dec 2015)
Oldest children from K2
(Jan-Mar 2016)
Outcome of interest
Oldest children from K3
Youngest children from K2
School cut-off
(January 1st)
Date-of-birth


Schooling group
(born from Oct-Dec 2015)
Non-schooling group
(born from Jan-Mar 2016)
Pretest
Attended 3rd year
of *preschool*
Attended 2nd year
of *preschool*
Posttest
Attended first grade
of *primary school*
Attended 3rd year
of *preschool*

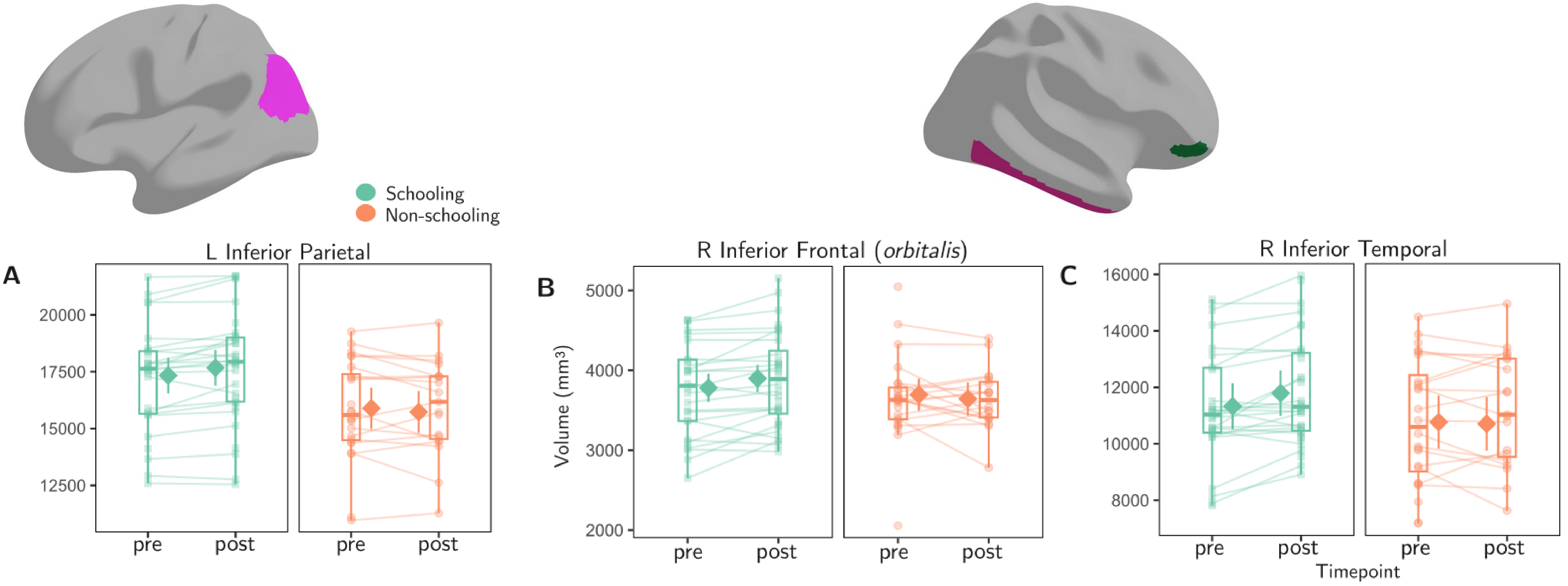

Schooling
Non-schooling
pretest
outcome parameter
outcome parameter
posttest
outcome parameter
outcome parameter
>

# Results: demographics

| Variables | Group | | | | *p* |
|---|---|---|---|---|---|
| | Non-schooling | | Schooling | | |
| | *n* | Summary statistic [a] | *n* | Summary statistic [a] | |
| Sex (female/male) | 28 | 19/9 | 36 | 19/17 | .223[e] |
| Socioeconomic status [b] Very low/low/middle/high | 28 | 0/2/11/15 | 36 | 1/8/15/12 | .202[e] |
| Head motion at pretest [g] None/mild/moderate/severe | 28 | 1/14/9/4 | 35 | 3/11/12/9 | .396[e] |
| Head motion at posttest [g] None/mild/moderate/severe | 21 | 1/10/8/2 | 35 | 3/18/11/3 | .921[e] |
| Age at pretest (months) | 28 | 66 (64 – 67) | 36 | 68.50 (65 – 70) | < .001[c] |
| Age at posttest (months) | 23 | 78 (76 – 79) | 35 | 81 (77 – 82) | < .001[c] |
| Verbal ability at pretest | 27 | 28 (18 – 36) | 35 | 31 (19 – 38) | 0.304 |
| Spatial ability at pretest | 27 | 15 (9 – 22) | 35 | 16 (8 – 26) | 0.178 |

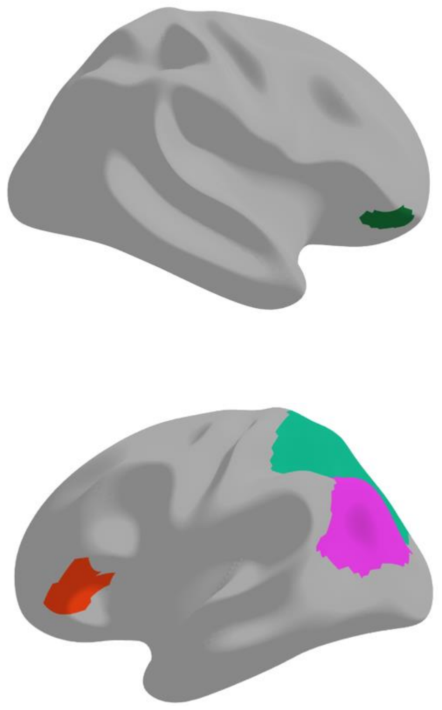
# Results: structural parameters

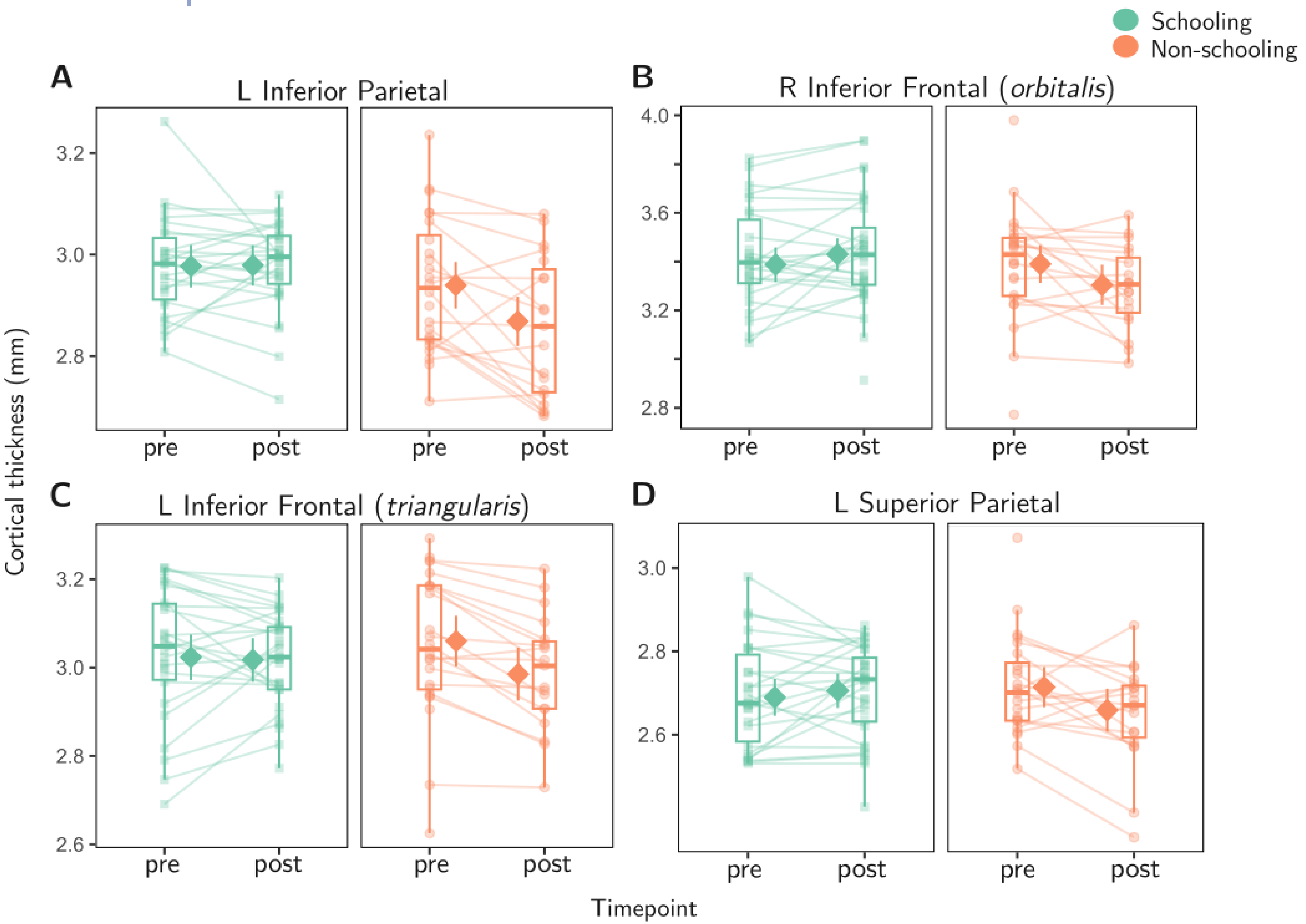
## Volume



# Results: structural parameters

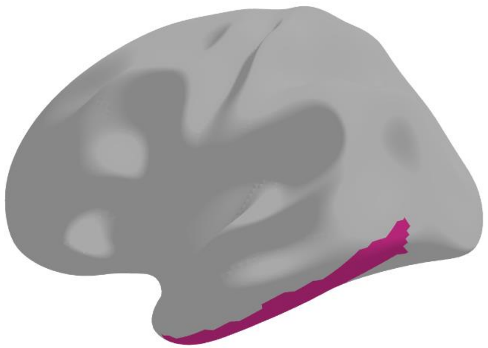
## Thickness

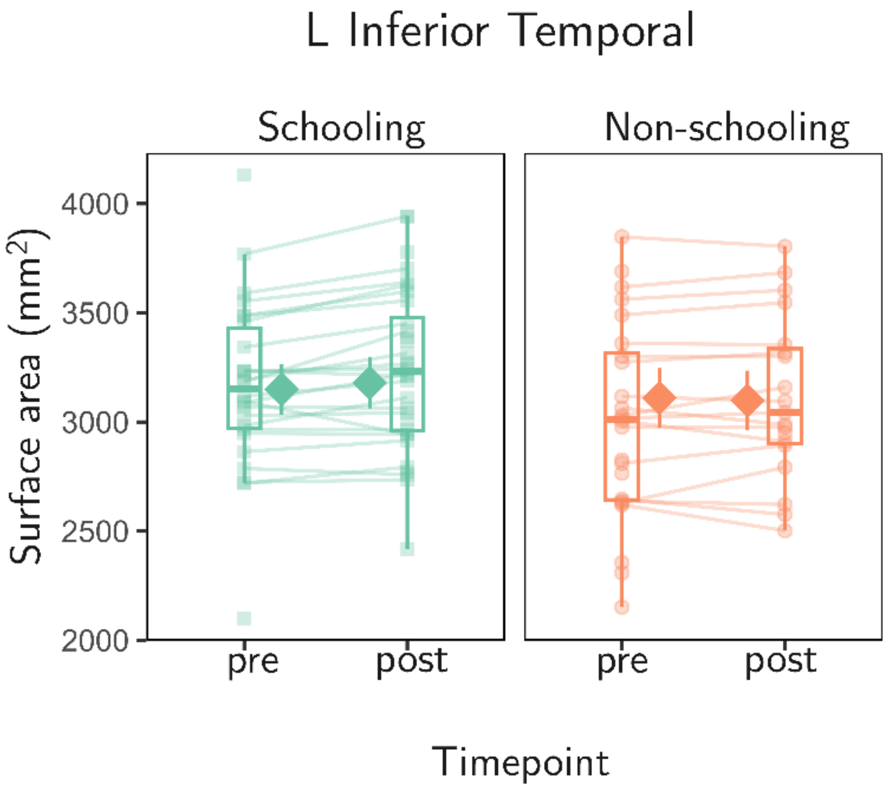




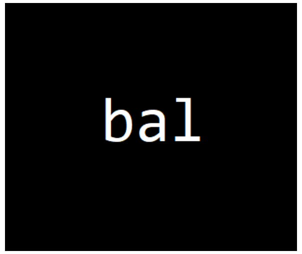
# Results: structural parameters

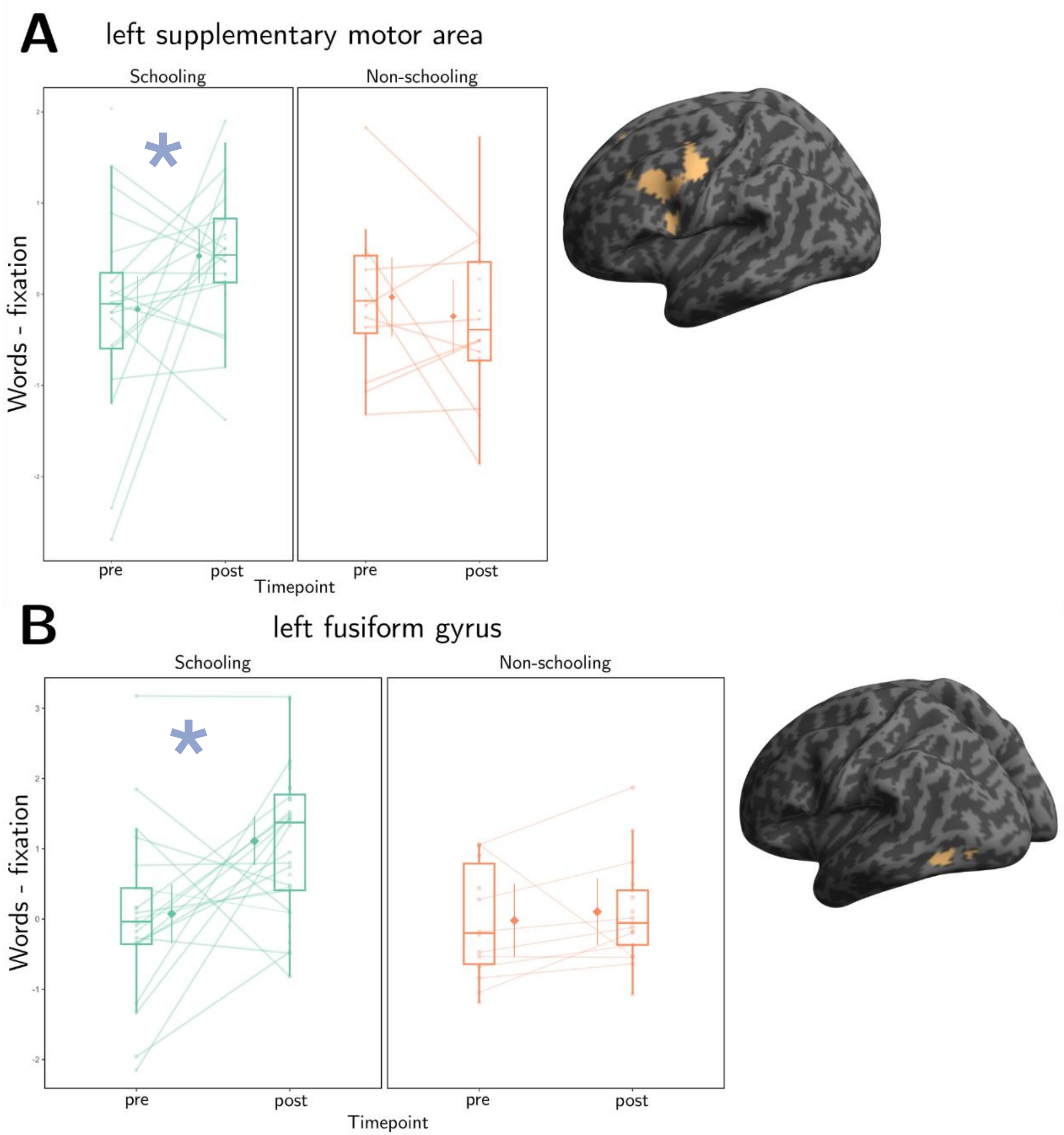
## Surface area



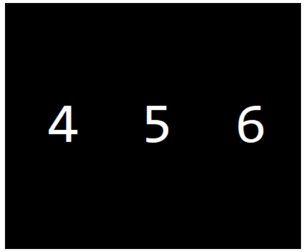


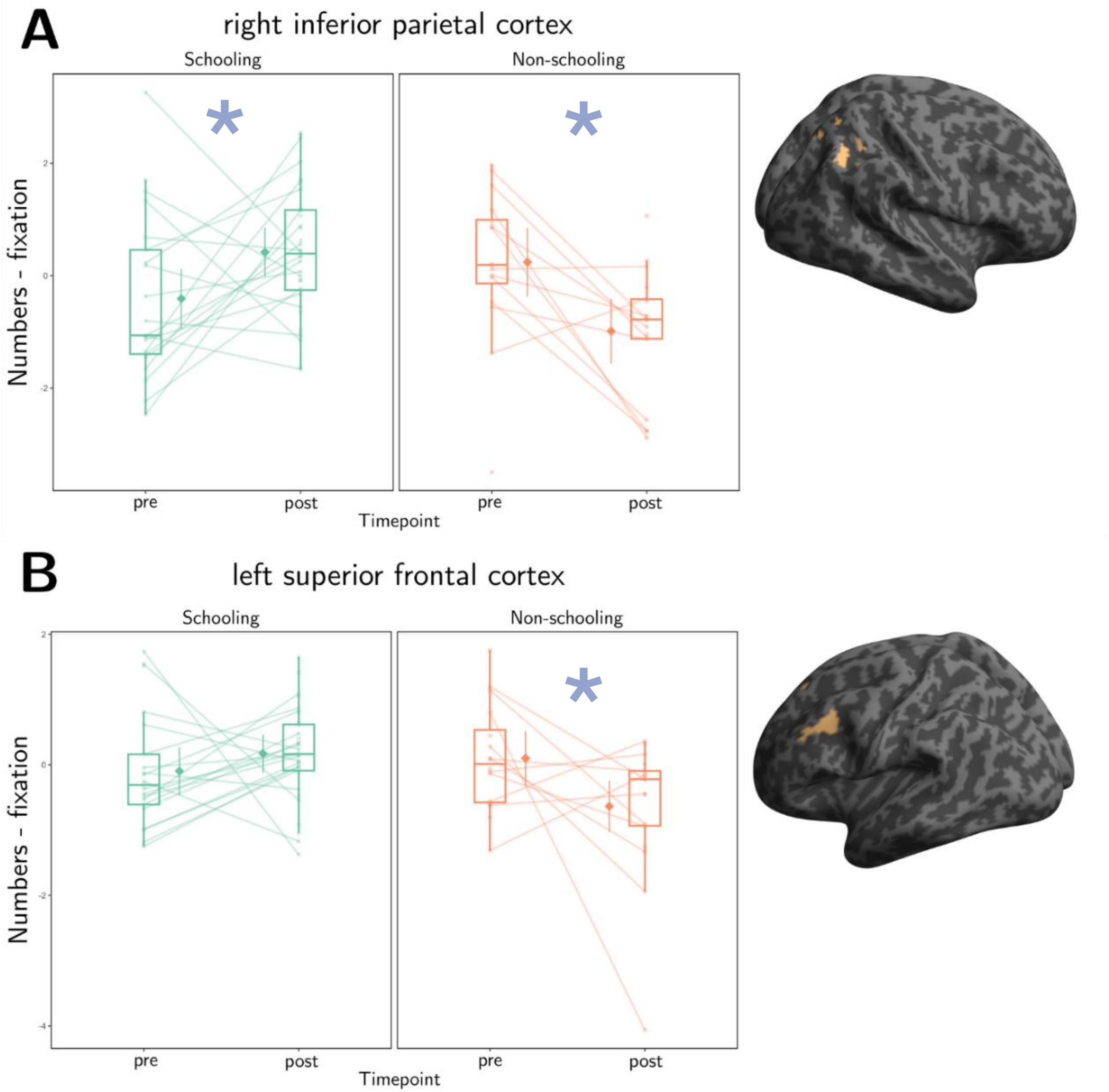
# Results: fmri
# Words - Fixation



# Results: fmri
# Numbers - Fixation



# Discussion

## Structural findings:

Most pronounced schooling effects on **volume and thickness**, to a lesser extent on surface area *(Jha et al., 2019)*

Schooling effects were observed as **increased volume and increased surface area**, but **stability in thickness** *(Walhovd et al., 2016; Tamnes et al., 2017; Ducharme et al., 2016)*

## Functional findings:

Increased activation for **numbers in the right IPC**, and increased activation for **words in left FG and left SMA**

**Schooling drives neural specialization** for words and numbers, over and above age *(Dehaene & Cohen, 2007)*

## ! But also:

**Interindividual variability** is meaningful and should not be interpreted as an irrelevant deviation from the mean *(Forkel et al., 2022)*

SCAN ME

Preprint fMRI results

(recently accepted)

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Check for updates

Experience-dependent cortical plasticity in response to formal schooling: Effects on networks for reading and mathematics

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# THANK YOU

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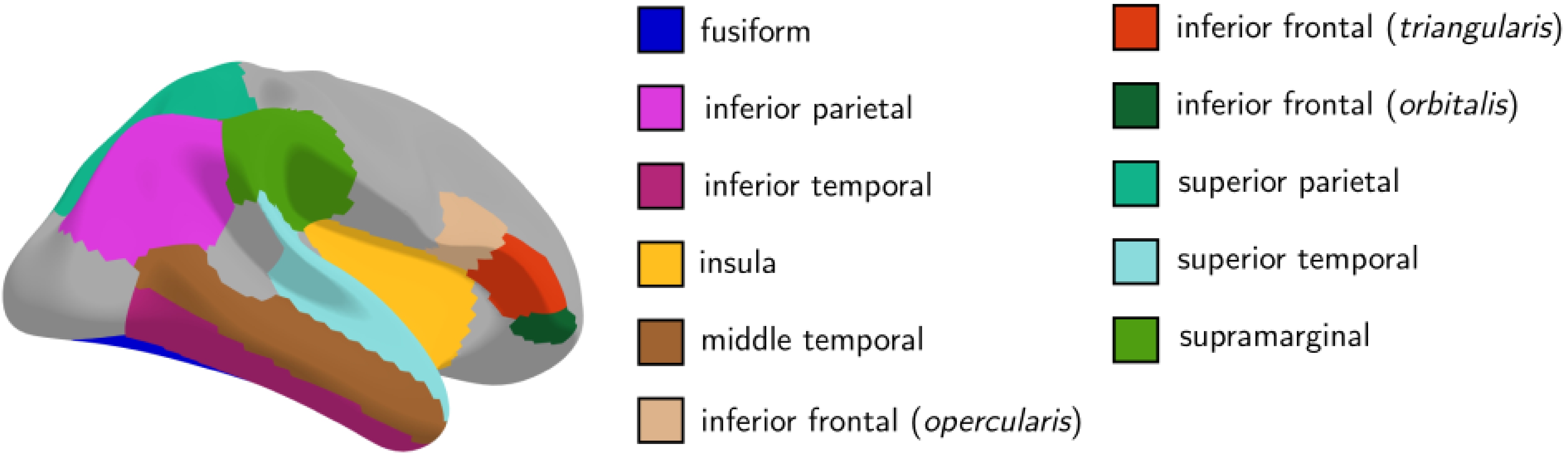
fusiform
inferior parietal
inferior temporal
insula
middle temporal
inferior frontal (*opercularis*)
inferior frontal (*triangularis*)
inferior frontal (*orbitalis*)
superior parietal
superior temporal
supramarginal

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Developmental Science WILEY

REGISTERED REPORT

# The role of formal schooling in the development of children's reading and arithmetic white matter networks

Floor Vandecruys[1,2] | Maaike Vandermosten[2,3] | Bert De Smedt[1,2]

AMERICAN PSYCHOLOGICAL ASSOCIATION
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# Education as a Natural Experiment: The Effect of Schooling on Early Mathematical and Reading Abilities and Their Precursors

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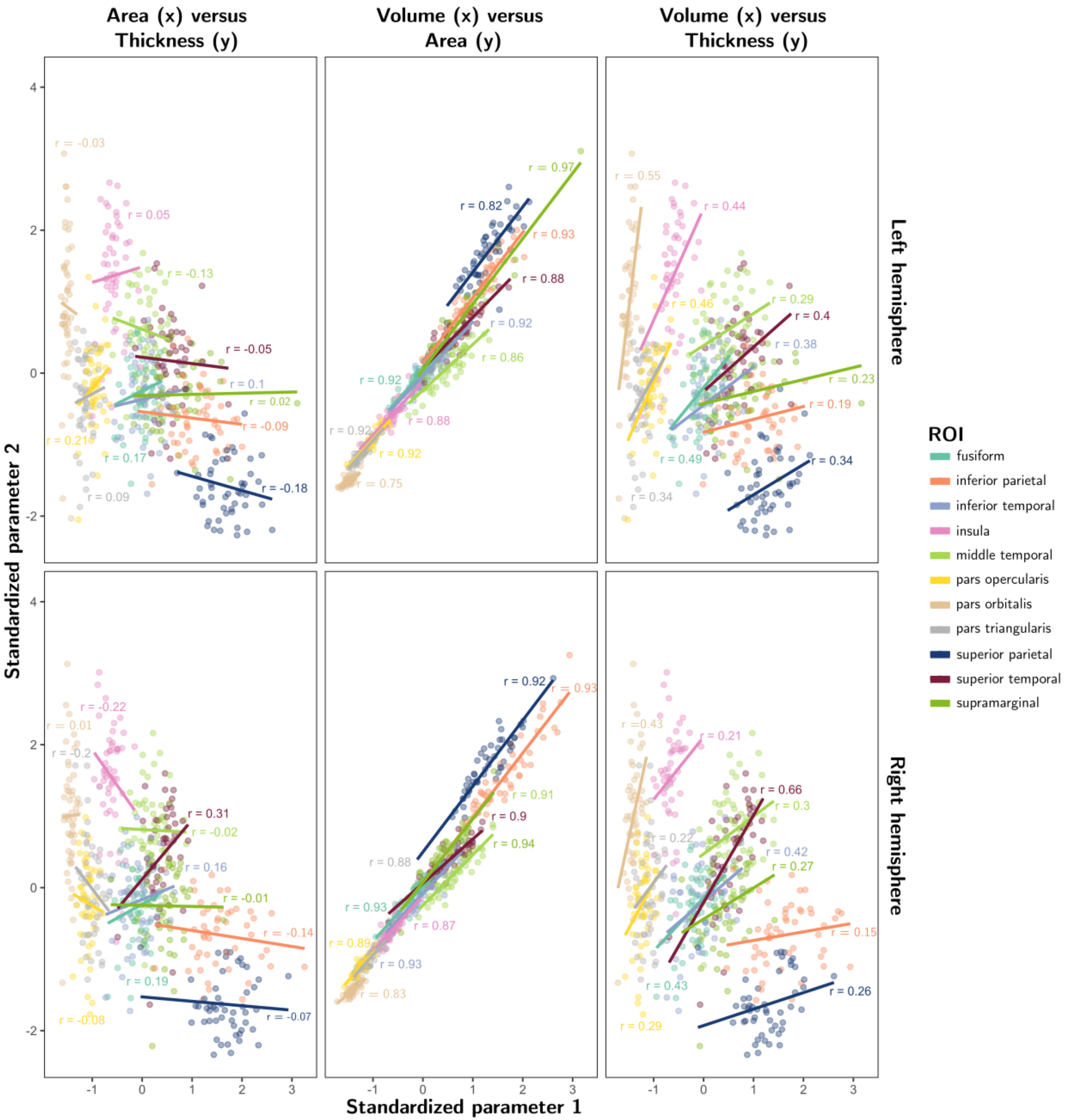
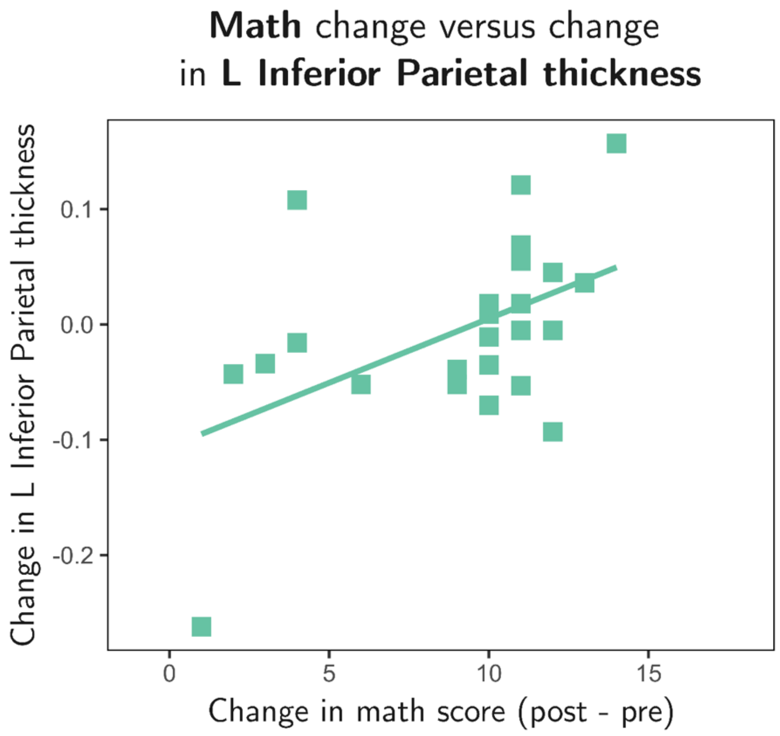


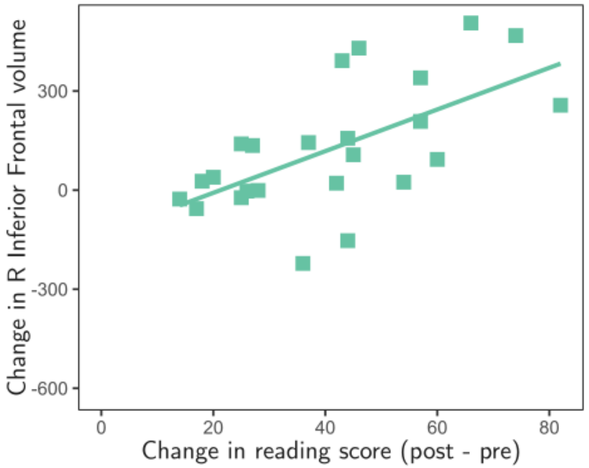
Area (x) versus Thickness (y)
Volume (x) versus Area (y)
Volume (x) versus Thickness (y)
Left hemisphere
Right hemisphere
Standardized parameter 1
Standardized parameter 2
ROI
fusiform
inferior parietal
inferior temporal
insula
middle temporal
pars opercularis
pars orbitalis
pars triangularis
superior parietal
superior temporal
supramarginal

# Results: correlations within schooling group with **mathematics**

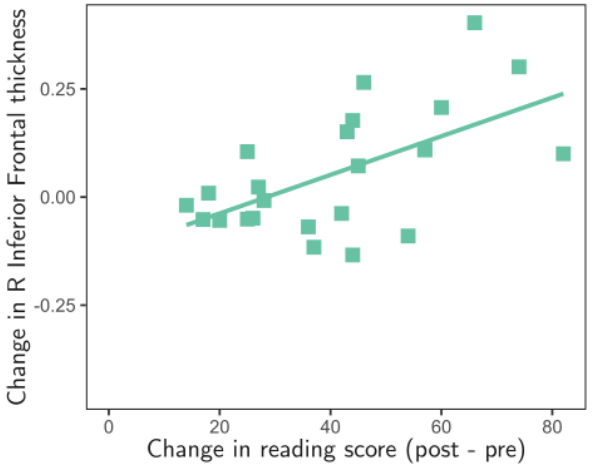


# Results: correlations within schooling group with **reading**

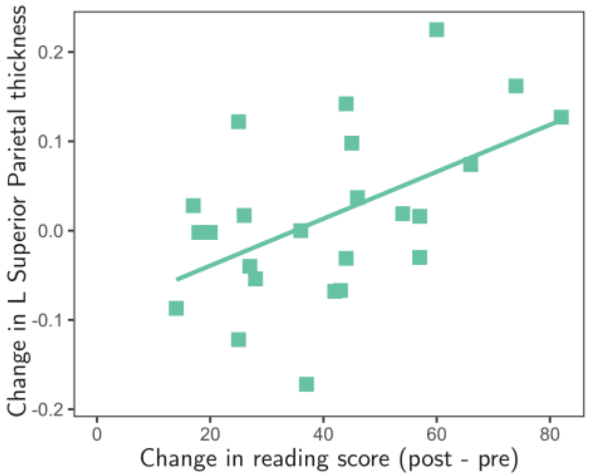
**Reading** change versus change in **R Inferior Frontal (*orbitalis*) volume**



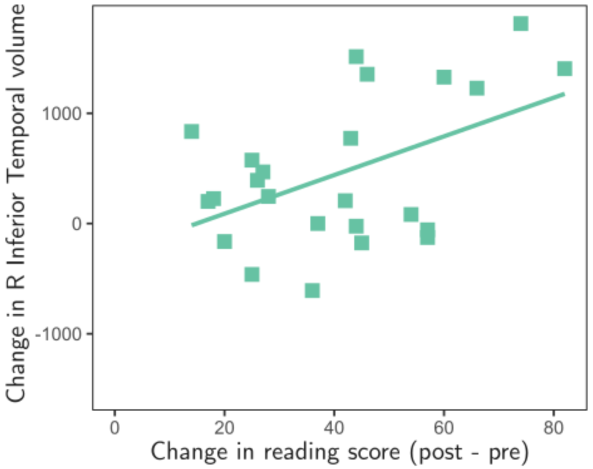

**Reading** change versus change in **R Inferior Frontal (*orbitalis*) thickness**




**Reading** change versus change in **L Superior Parietal thickness**




**Reading** change versus change in **R Inferior Temporal volume**



# fMRI Results: correlations within schooling group with **reading**